

RESEARCH ARTICLE

Ambient IoT Toward 6G: Standardization, Potentials, and Challenges

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ABSTRACT The Ambient Internet of Things (A-IoT) has emerged as a critical direction for achieving effective connectivity as the IoT system evolves to 6G. However, the introduction of A-IoT technologies, particularly involving backscatter modulation, poses numerous challenges for system design and network operations. This paper surveys current standardization efforts, highlights potential challenges, and explores future directions for A-IoT. It begins with a comprehensive overview of ongoing standardization initiatives by the 3rd Generation Partnership Project (3GPP) on A-IoT, providing a solid foundation for further technical research in both industry and academia. Building upon this groundwork, the paper conducts an analysis of critical enabling technologies for A-IoT. Moreover, a comprehensive A-IoT demonstration system is designed to showcase the practical viability and efficiency of A-IoT techniques, supported by field experiments. We finally address ongoing challenges associated with A-IoT technologies, providing valuable insights for future research endeavors.

INDEX TERMS Internet of things (IoT), backscatter communications, 6G.

I. INTRODUCTION

Recently, Internet of things (IoT) has experienced rapid growth, with the global number of IoT devices reaching 13.2 billion in 2022. Meanwhile, it is expected to exceed 34.7 billion by 2028 as traditional industries undergo intelligent transformations [1]. IoT has revolutionized connectivity, allowing all things to be interconnected intelligently through advanced technologies like artificial intelligence (AI). Wireless communication technologies such as Radio Frequency Identification (RFID), Zigbee, cellular mobile communications, and others can provide connectivity services for IoT applications.

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Specially, cellular-based IoT networks gained widespread acceptance around the world with 2G and 3G connectivity two decades ago. Nowadays, the introduction of 4G and 5G cellular networks provides benefits such as larger bandwidth, lower latency, and increased capacity, making them preferred choices for emerging IoT applications [2], [3]. For example, narrowband IoT (NB-IoT) based on 4G networks effectively meets the requirements of various low data speed IoT applications [4].

Unlike 4G, 5G introduces significant changes in both architecture and capabilities, particularly in its support for massive IoT deployments. In addition to NB-IoT and LTE-M technologies, 5G RedCap (Reduced Capability) was developed to cater to various IoT applications with features such as low RF complexity and cost-efficiency. However,

these existing IoT technologies are insufficient to meet the future demands of massive IoT, especially regarding ultra-low power consumption. There is an urgent need for advancements in cellular-based IoT networks as we move toward 6G [5], which is expected to offer enhanced and extended 5G capabilities in terms of achievable data rates, latency, and so on [6]. The 6G vision aims to establish the foundation for a seamless cyber-physical world, requiring significant technological advancements, particularly in IoT [7].

Compared to 5G, 6G technology promises to meet more stringent IoT requirements, including massive connectivity, Ultra-Reliable Low-Latency IoT Communications, improved communication protocols, extended IoT network coverage, and smart IoT devices [8]. Consequently, powering billions of IoT devices becomes one of significant challenges in 6G IoT. Traditional solutions, such as deploying power cables or regularly replacing/charging batteries, are impractical at this scale. In contrast, ambient communications utilizing backscatter modulation technology have the potential to establish long-term, maintenance-free, and energy-efficient networks due to the ultra-low power consumption characteristics [9]. As a result, both industry and academia are increasingly focusing on IoT networks based on ambient communications, also known as Ambient IoT (A-IoT). A-IoT refers to an ecosystem of numerous interconnected objects, each linked to a network through low-cost devices that utilize backscatter communications. A-IoT devices typically harvest energy from their surroundings, such as Radio Frequency (RF) signals, solar energy, vibration, or heat. A-IoT is not intended to replace existing 5G IoT technologies but is designed to extend the capabilities of the 6G system in areas where 5G falls short.

Recently there are numerous studies on ambient IoT (A-IoT) such as ambient backscatter modulation and so on [10], [11], and [12]. However, the majority of these studies solely focused on individual technique without taking into account the overall design and operations from the system view [13]. Fortunately, the 3rd Generation Partnership Project (3GPP) has recently begun to pay attentions to A-IoT, and has initiated systematic discussions and standardisation efforts in related technologies [14], [15]. The efforts will undoubtedly accelerate the development of A-IoT systems related to cellular mobile communications. It should be noted that, however, the studies and standardisation of key technologies in A-IoT systems will face a lot of challenges and issues. This paper aims to analyze and discuss them in an exploratory manner. The main contributions of this paper are as follows:

- 1) We thoroughly identify and examine the current state of A-IoT systems within the technical framework of the 3GPP. The analysis encompasses user cases, network topologies adopted by 3GPP for integrating backscatter communications into A-IoT systems. The aim is to provide a comprehensive understanding of the existing standardization and open the door for more in-depth research.

- 2) A detailed exploration is conducted on the traits of key techniques in A-IoT systems. It involves the ones that are essential to the operation of A-IoT systems, such as backscatter modulation, interference cancellation/avoidance, and multiple access methods.
- 3) We design and implement an experimental platform tailored explicitly for A-IoT systems. This platform can serve as a practical demonstration, showcasing the feasibility and effectiveness of the A-IoT techniques. Particularly, field experimental results are presented in terms of error rate performances under typical indoor environments.
- 4) This paper also tentatively presents the challenges related to system operation and performance enhancement that A-IoT systems need to tackle in the next steps.

This paper is organized as follows. A detailed description on current progress of 3GPP standardization on A-IoT is provided in Section II. Next, we discuss key techniques conceived for A-IoT systems in Section III. We then present our demonstration system and its experimental results in Section IV, and also discuss some open issues in Section V. Finally, our conclusions are given in Section VI.

II. AMBIENT IoT STANDARDIZATION IN 3GPP

The motivation behind the A-IoT study in 3GPP is to facilitate the use of ultra-low cost and ultra-low power devices for IoT applications. Currently, study item on A-IoT in 3GPP has been finished. This study specifically concentrates on IoT technologies, which is suitable for deployment in sixth generation (6G). An agreement has been reached among companies regarding representative Use Cases (rUCs), connectivity topologies and deployment scenarios as outlined in [14]. However, there are still numerous open issues that need to be solved in the stage of work item in 3GPP [16].

A. REPRESENTATIVE USE CASES

To facilitate requirement definitions, 3GPP has established two sets or levels of grouping use cases of A-IoT systems [14], i.e.,

- **Grouping A:** This grouping categorizes use cases based on the deployment environment, distinguishing between *indoor* and *outdoor* scenarios.
- **Grouping B:** This grouping classifies use cases based on functionality and application, including *inventory management*, *sensor applications*, *positioning*, and *command applications*.

These two groupings have led to the eight kinds of rUCs, as outlined in Table 1. These rUCs serve as examples representing different application scenarios of A-IoT.

The deployment scenarios for A-IoT rUCs are expected to be represented by a set of characteristics, including environment, base station characteristics, connectivity topology, spectrum, co-existence with existing 3GPP systems, and traffic assumption. As a result, such grouping results may facilitate the radio access network (RAN) design and implementation solutions that are tailored to the specific

TABLE 1. Representative use cases in 3GPP RAN.

Group B Group A	Inventory	Sensor	Positioning	Command
Indoor	rUC #1: - Automated warehousing - Automobile manufacturing - Smart laundry - Same with rUC5	rUC #2: - Smart homes - Base station machine room environmental supervision - Smart agriculture	rUC #3: - Ranging in a home - Positioning in shopping centre - Museum Guide - Same with rUC7	rUC #4: - Device Permanent Deactivation - Electronic shelf label - Same with rUC8
Outdoor	rUC #5: - Medical instruments inventory management and positioning - Non-public network for logistics - Airport terminal/shipping port - Automated supply chain distribution	rUC #6: - Smart grids - Forest Fire Monitoring - Dairy farming - Smart bridge health monitoring	rUC #7: - Finding remote lost item - Location service - Personal belongings finding	rUC #8: - Online modification of medical instruments status - Device activation and deactivation - Elderly Health Care - Controller in smart agriculture

requirements and characteristics of a specific deployment scenario.

On the other hand, the practical deployment performance of specific user cases is heavily dependent on the capability of A-IoT devices, which are quite different from traditional user equipment (UE). As a result, device categorization needs to be defined based on corresponding characteristics such as energy storage capacity and the ability to generate RF signals for transmission. There might be three device categories for A-IoT applications, i.e.,

- **Device A:** No energy storage, and no independent signal generation/amplification, i.e., backscattering transmission.
- **Device B:** Having energy storage, and no independent signal generation, i.e., backscattering transmission. The amplification for reflected signals can be included based on the stored energy.
- **Device C:** Having energy storage and independent signal generation, i.e., active RF components for transmission.

All the devices are expected to have low power consumption. The capabilities of different device types can vary significantly, especially between Device A and Device C. Typically, Device A or B can be powered by an RF energy harvester at the micro-watt (μW) or sub-micro-watt (sub- μW) level, allowing for a transmission range of a few tens of meters. However, Device C may not be suitable for RF energy as a power supply. One critical issue is that the output power of the RF energy harvester may be even lower than the self-discharging power of the super capacitor used by Device C, which is typically a few micro-watts (μW) or higher.

Since the uplink transmission is based on backscattering, Devices A and B may not be suitable for use cases involving device-originated traffic. In contrast, Device C, with the

ability to generate uplink RF signals on its own, is assumed to support both device-terminated and device-originated traffic.

B. CONNECTIVITY TOPOLOGIES

The connectivity topologies are critical in deployment scenarios. In Fig. 1, four connectivity topologies for A-IoT networks have been defined and will be studied further in 3GPP. In each of these topologies, the A-IoT device can receive a carrier wave, which is used to energize the device, from other nodes within or outside the topology. The links in these topologies can be either bidirectional or unidirectional, depending on the specific configuration and requirements of the network. There are two types of links in A-IoT systems, i.e., “Uu” and “Ua” links. The former denotes the air interface connecting traditional UE to the RAN. The latter is newly designed for the connection of A-IoT devices. The nodes involved in these topologies include the base station (BS), assisting node, intermediate node and UE. The placement of these nodes can be a combination of indoor and outdoor locations, allowing for flexibility in network implementation. Different topologies have distinct characteristics and suitable scenarios. Next, we briefly discuss four typical topologies.

1) TOPOLOGY 1 (BS $\leftrightarrow$ A-IoT DEVICE)

In this simplest topology, the A-IoT device establishes a direct bidirectional communication link with a BS using a new air interface, i.e., “Ua”. This air interface is responsible for transmitting A-IoT data and signalling. In those scenarios where high service availability is required over a large local or wide area, it is challenging for Device A and B to directly connect to outdoor macro- or micro-cell BSs. Device C is expected to be more suitable for ensuring continuous outdoor coverage, albeit at the expense of higher power consumption for transmission. Compared with Device A and B, Device C

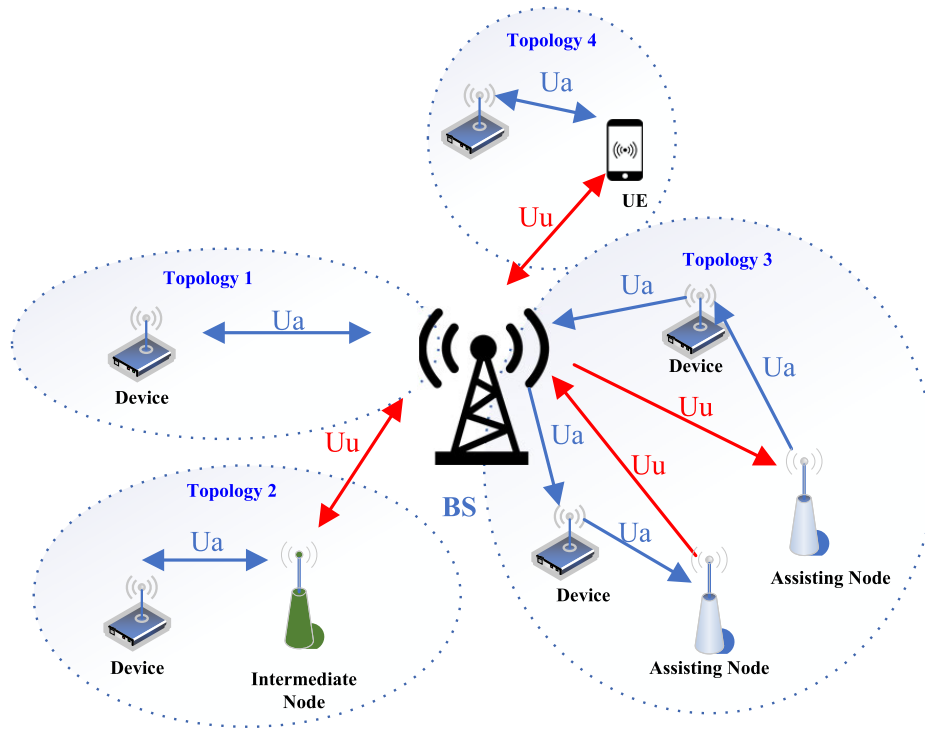


FIGURE 1. Typical connectivity topologies for A-IoT networks defined in 3GPP.

has ability to maintain a reliable connection with outdoor BSs while operating efficiently over a large area.

2) TOPOLOGY 2 (BS $\leftrightarrow$ INTERMEDIATE NODE $\leftrightarrow$ A-IoT DEVICE)

In this topology, the bidirectional communication link between the A-IoT device and the BS is facilitated through an intermediate node. This intermediate node can be various kinds of equipments, such as a relay node, integrated access/backhaul node, UE, repeater, and so on, only requiring limited full duplex capability. The role of the intermediate node is to transfer information between the BS and the A-IoT device, acting as a bridge in the communication process.

From the perspective of the device, this topology may have similarities to Topology 1. However, it offers the advantage of providing larger coverage. By leveraging the intermediate node, the communication range and coverage area of the A-IoT system can be extended, enhancing the connectivity options.

3) TOPOLOGY 3 (BS $\leftrightarrow$ ASSISTING NODE $\leftrightarrow$ A-IoT DEVICE $\leftrightarrow$ BS)

This topology is specifically designed to address the use case where there is an imbalance in the communication range between the downlink and uplink. To overcome this imbalance, an assisting node is introduced to relay data between the BS and the A-IoT device. In Topology 3, the communication scenarios can vary, i.e.,

- The A-IoT device can transmit data or signalling to the BS while receiving data or signalling from the assisting node.
- Alternatively, the A-IoT device can receive data or signalling from the BS while transmitting data or signalling to the assisting node.

The specific categories of the assisting node are similar with those of the intermediate node. Both the BS and assisting node do not require full duplex capability in this topology. On the other hand, the A-IoT device such as Device A or B has the option to receive a carrier wave from another node within or outside the topology. This flexibility allows for efficient signal energization of the A-IoT device.

4) TOPOLOGY 4 (UE $\leftrightarrow$ A-IoT DEVICE)

In this topology, the bidirectional communication link between the A-IoT device and a UE takes place without the involvement of the BS. The communication includes A-IoT data and signalling. Moreover, it might work on an unlicensed band where the UE does not need to establish a connection via the “Uu” link. However, operating in an unlicensed band imposes certain limitations, such as lower transmit power resulting in less coverage compared to licensed band. Despite these limitations, this topology is still useful, particularly for indoor environment where the coverage area is small.

III. POTENTIAL TECHNOLOGIES IN A-IoT

Within the technical framework established by 3GPP, the development and maturation of various A-IoT related

technologies are essential prerequisites for realizing a functional A-IoT system during the 6G era. Despite the emergence of diverse ambient communication technologies in recent years, this section only concentrates on potential technologies that might be applied in the A-IoT system.

A. BACKSCATTER MODULATION

For simplicity, the incoming signal is defined as a sine wave with the carrier of f_{in} and received at the A-IoT device, i.e.,

$$S_{in}(t) = A_{in} \exp[j(2\pi f_{in}t + \theta_{in})]. \quad (1)$$

where A_{in} and θ_{in} are the amplitude and phase of sine wave, respectively. Afterwards, the received signal is backscattered at the antenna of the A-IoT device, resulting in the product of the received signal and the backscattering coefficient $\Gamma(t)$, i.e.,

$$\begin{aligned} S_{out}(t) &= \Gamma(t)S_{in}(t) \\ &= |\Gamma(t)|A_{in} \exp\{j[2\pi f_{in}t + \theta_{in} + \theta_{\Gamma}(t)]\}. \end{aligned} \quad (2)$$

where $\Gamma(t) = |\Gamma(t)|e^{j\theta_{\Gamma}(t)}$. As depicted in (2), the desired modulation scheme can be achieved by adjusting the amplitude, and phase of the backscattering coefficient. The fundamental concept behind backscatter modulation is to manipulate the antenna impedance Z_a at the device, thereby generating the backscattering coefficient Γ , i.e.,

$$\Gamma(t) = \begin{cases} \frac{Z_m(t) - Z_a^*}{Z_m(t) + Z_a^*}, & m = 1, 2, \dots, M, \\ 0, & m = 0, \end{cases} \quad (3)$$

where $Z_m(t)$ is the load impedance and $*$ is the complex conjugate operator. As shown in (3), there are two modes of the device, i.e., active state ($m = 1, 2, \dots, M$) and inactive state ($m = 0$). In the active mode, there are M switch states available. Consequently, various backscatter modulation techniques have been developed, including the Amplitude-Shift Keying (ASK), frequency-shift keying (FSK), Phase-Shift Keying (PSK), and Quadrature Amplitude Modulation (QAM). However, the two-state modulation is commonly employed due to its simplicity. For instance, the On-Off Keying (OOK) and Binary PSK (BPSK) modulation are widely used by RFID tags, serving as the simplest modulation schemes [17].

The modulated signals backscattered from the A-IoT device need to be reliably detected at the receiver. In the existing literature, a few detection techniques have been already developed [18], [19]. Owing to the feature of simplicity, non-coherent detection, such as the envelop detector, is the most widely used. However, it is only appropriate for ASK and FSK modulations with low data rate that do not require carrier phase information. By contrast, in those use cases where high data rates are required, coherent detection with PSK modulation is preferred, but at the cost of pilot overhead and implementation complexity.

It is highly likely that BPSK, QPSK, and even 16QAM may be adopted as optional modulation schemes in A-IoT systems.

Nevertheless, given the requirement for large transmission distances in most of rUCs, either BPSK or QPSK is preferred to be used with the coherent detection.

B. INTERFERENCE CANCELLATION/AVOIDANCE

The performance of an A-IoT system is often limited by short communication ranges and low data rates. These limitations arise from the presence of Direct Path Interference (DPI) and weak backscatter signals. In a traditional backscatter system, a continuous unmodulated sinusoidal wave is typically employed as the source signal, which can be easily identified by a receiver. Consequently, the non-coherent or coherent receivers can be conveniently implemented for detecting the modulated backscatter signals.

However, these detection techniques cannot be directly applied in A-IoT systems. This is because these systems typically employ an unknown modulated signal from a specific legal system as the source signal, which may introduce strong DPI to the backscatter signal. As a result, new techniques and strategies need to be developed for A-IoT systems. Specially, DPI cancellation or avoidance methods have garnered attention in 3GPP and some of them may be feasible in typical deployment scenarios, i.e.,

- 1) **Interference Cancellation:** To mitigate the DPI, a technique such as Successive Interference Cancellation (SIC) can be employed [10]. Specifically, SIC involves jointly decoding the ambient and backscatter signals when the Channel State Information (CSI) is known at the receiver. By leveraging the knowledge of the CSI, the receiver first detects the RF-source signal, then subtracts its resultant direct-link interference from the received signal, and recovers the ambient backscatter signal. Finally, based on the recovered ambient backscatter signal, the receiver re-estimates the RF-source signal. This technique enables more accurate and reliable decoding of the backscatter signals in the presence of strong DPI.
- 2) **Interference Avoidance:** One approach to deal with DPI in backscatter systems is by shifting the frequency of the backscatter signal to an adjacent non-overlapping frequency band or a guard band [11]. To this end, potential DPI can be excluded through filtering, effectively reducing its impact on the backscatter signal. This frequency shift technique can improve the received Signal-to-Noise Ratio (SNR) and enhance the reliability of the backscatter communications. Another strategy is to exploit spatial diversity to separate the backscatter signal from the direct signal. This allows the receiver to differentiate between the two signals, even without relying on a special ambient signal or having prior knowledge of the CSI.

C. MULTIPLE ACCESS

The selection of multiple access methods has a significant impact on the reliability and efficiency of A-IoT systems. The

TABLE 2. Comparison of different multiple access methods.

	OFDM Modulation	Backscatter Modulation
FDMA	- Simply assign subsets of OFDM subcarriers to individual users, e.g., OFDMA - Few users served simultaneously	- Assign different frequencies to individual devices - Simple but less flexible; and limited by the available bandwidth
TDMA	- Allow multiple users to share OFDM subcarriers by dividing the signal into different time slots - Strict synchronization requirement and high overhead	- Different devices are assigned different time slots either deterministically or dynamically - Very popular multiplexing method for backscatter communications
CDMA	- Users share either different subcarriers or time slot by using different spreading codes - Various methods to combine CDMA with OFDM, e.g., Multi-carrier CDMA; usually with high implementation complexity	- Devices spread the information using different orthogonal sequences and then convey the spread information by reflecting and modulating - Robust against the noises but high sensitivity to time synchronization offset
NOMA	- Multiple users can be allocated on a subcarrier at the same time by adopting NOMA with OFDM - High spectral efficiency with the cost of the implementation complexity and overhead	- Devices in different spatial regions are multiplexed to implement PD-NOMA - Limited by the implementation feasibility and power consumption

majority of current studies on backscatter communications employs centralised multiple access methods, such as Time-Division Multiple Access (TDMA) and Frequency-Division Multiple Access (FDMA), in which a central entity allocates specific resources to multiple devices in the system. FDMA is simple but has its limitation. For instance, a BS assigns different frequencies to all A-IoT devices in a FDMA system. The devices should be able to flexibly adjust their transmission frequencies within the entire system band. However, this raises the cost of individual device. Additionally, the available system bandwidth is typically limited. Both of these factors render FDMA-based systems unsuitable for large-scale deployments. On the other hand, TDMA is a more commonly used in backscatter communication systems. Unfortunately, achieving high-precision time synchronisation among devices is a huge burden for low-cost and low-power A-IoT devices. As a result, new multiple access methods need to be developed specifically in A-IoT systems. For example, a Coded-Backscatter Multiple Access (CBMA) scheme was proposed for backscatter communication [20]. It mainly consists of a correlation-based detector and power control at the device. The former aims to reduce the negative effects caused by the asynchronous problem, while the purpose of the latter is to improve the performance against the significant power difference among devices. Furthermore, A-IoT systems are expected to support a large number of devices simultaneously. Therefore, it is necessary to investigate advanced multiple access methods with efficient concurrent transmissions. Power-domain Non-Orthogonal Multiple Access (NOMA) is one of the methods

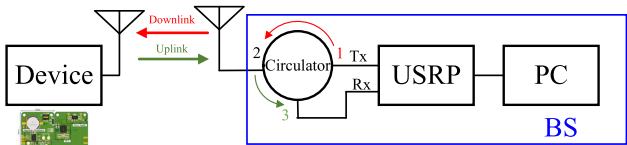


FIGURE 2. Illustrating of an A-IoT demonstration system.

that can achieve this goal [21]. The A-IoT devices in different spatial regions are multiplexed to implement NOMA. Unlike conventional mobile devices that can actively adjust their transmit power, the reflection coefficients for multiplexed A-IoT devices must be set to different values to more effectively leverage the benefits of power-domain NOMA. Different multiple access methods can be applied in either OFDM or backscatter modulation schemes, which are compared and analyzed in Table 2. To conclude, striking a balance of between performance and complexity, A-IoT is likely to utilize TDMA as its primary multiple access method. To enhance support for applications with a greater number of devices or higher transmission rates, there is a need to explore

TABLE 3. Main parameters of A-IoT demonstration system.

Parameter Description	Value
Carrier Frequency	925 MHz
Bandwidth	1 MHz
Modulation	BPSK / OOK
BS Transmit Power	20 dBm
BS Antenna Gain	5 dBi
Device Antenna Gain	2 dBi
Maximum Isolation of Circulator	20 dB
Insertion Loss of Circulator	0.4 dB

optional multiple access methods, including those based on CBMA, NOMA, and so on.

IV. IMPLEMENTATION AND DEMONSTRATION OF A A-IoT PROTOTYPE SYSTEM

Although A-IoT technologies are still in development, it is still very necessary to build a practical platform to demonstrate the feasibility of A-IoT systems. From a technical perspective, the backscatter communication used in future A-IoT systems will likely follow principles similar to those used in traditional RFID. We focus on showing the feasibility of using backscatter communication to achieve long-distance communication within the current 3GPP framework. As shown in Fig. 2, our demonstration system comprises of a BS and an A-IoT device (also known as a tag). In this setup, only one device was considered. We plan to include more devices, which will allow us to address challenges related to network topology in the near future. The BS consists of a general-purpose computer, a commercial Universal Software Radio Peripheral (USRP) X310, and a polarized panel antenna with a gain of 5 dBi. The USRP Hardware Driver library is utilized to establish concurrent transmitting and receiving threads which are responsible for various kinds of signal processing procedures. The signals interact with the USRP X310 through Gigabit Ethernet, which handles the signal conversion between baseband and radio frequency. Meanwhile, we use a circulator¹ to separate the transmitted and received signals in the experiments. **Port** 1 and 3 of the circulator are connected to the USRP X310's transmitting and receiving ports, respectively, while **Port** 2 is connected to the antenna. Finally, A-IoT devices are developed by using an ARM processor and its hierarchical circuits on a four-layer printed circuit board with dimensions of 30 mm × 40 mm. The main parameters of the demonstration system are shown in Table 3.

Deriving a theoretical closed-form expression that directly relates BER to distance (or SNR) is challenging due to the effects of complex fading channels in real-world environments. Therefore, we conducted field experiments to evaluate

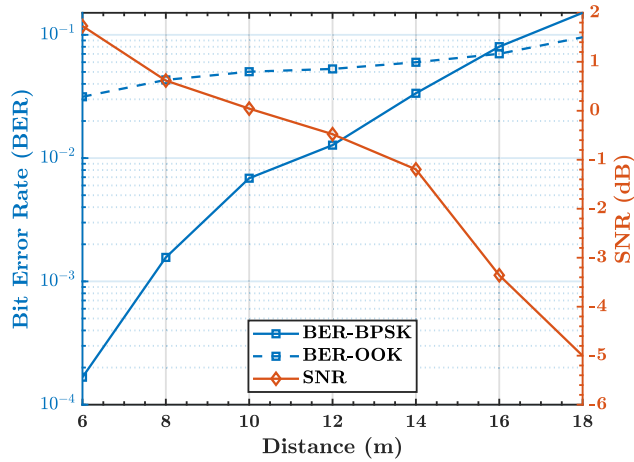
the performance of the A-IoT system in a typical indoor setting. In these experiments, we fixed the base station's location and moved the device (tag) to achieve varying communication distances. At the receiver end, specific algorithms were used to estimate different SNR values, and the corresponding BER and data rate were measured. We also implemented interference cancellation technique in our receiver to minimize the impact of interference on system performance, particularly over longer communication distances.

The communication process between the BS and the device adopts a “*BS first talk, then device response*” mechanism in our demonstration system. In **Phase 1**, the BS only sends a carrier signal to the device, which does not reflect it. Then, the power of the noise can be easily estimated by averaging the difference between the received signal and the carrier signal, i.e., P_n . In **Phase 2**: the BS continues to send carrier signals, which are then reflected by the device along with its data. Subsequently, the BS receives the reflected signal. Now, we can compute the sum of the power of the useful signal and the noise by averaging the received signal after subtracting the carrier signal, i.e., $P_2 = P_s + P_n$. Next, the SNR can be conveniently calculated by $\gamma = \frac{P_s}{P_n} = \frac{P_2 - P_n}{P_n}$.

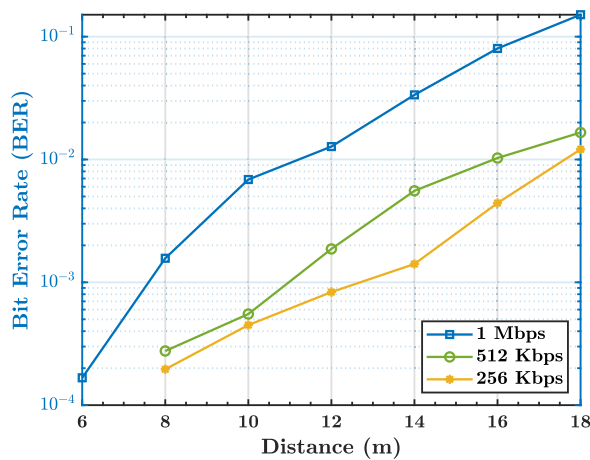
A customized frame structure for the data transmission is defined. Each frame consists of three parts, i.e., a 16-bit preamble, 992 bits of user data, and a 16-bit Cyclic Redundancy Check (CRC) bits. The preamble is utilized for frame synchronization and channel estimation. Then, estimated channel state information can be employed for coherent demodulation. The data part is used for transmitting the sensory data collected by the device, while the CRC bits provides data integrity verification. After receiving the reflected signal, the BS first performs frame synchronization by cross-correlating the known preamble with the received data to search for the frame header. Once the frame header is successfully detected, the BS then utilizes the preamble for channel estimation. Subsequently, the data part undergoes a series of modules including matched filtering, self-interference cancellation, and coherent demodulation to retrieve the transmitted data from the device.

As illustrated in Fig. 3(a), with a data rate of 1 Mbps using two modulation methods, i.e., OOK and BPSK, respectively, the received SNR decreases as the distance rises, resulting in a higher Bit Error Rate (BER). It is obvious that the BER of BPSK modulation achieves the better performance. However, the BER of BPSK becomes larger than that of OOK in case of low SNR, e.g., -5 dB. This is primarily because acquiring precise phase information becomes more difficulty due to its susceptibility to the noise compared with the amplitude information. Furthermore, Fig. 3(b) indicates that the BER performance improves when the data rate is reduced in cases of BPSK modulation. In other words, when the required BER performance is given, the communication distance can be extended by sacrificing the data rate in order to achieve the coverage requirement.

¹Here, the circulator model is QUEST Microwave SL8996C01, and the details of the circulator can be found at <http://www.questmw.com/>



(a) Different modulation schemes.



(b) Different data rates.

FIGURE 3. Measured BER and SNR versus the distance in an A-IoT demonstration system.

V. OPEN ISSUES AND FUTURE DIRECTIONS

A. MULTI-ANTENNA TRANSMISSION

One of the potential solutions to enhance the communication reliability and range of A-IoT systems is to apply the multi-antenna technologies. In general, it is more convenient to deploy multiple antennas at the BS, assisting node and intermediate node, where the coherent detector with multiple receiving antennas might be used to extend the communication distance.

Moreover, using multiple antennas on an A-IoT device can provide diversity gain and thus further improve detection performance. It has the potential to increase the IoT device's backscatter signal power and energy harvest. More importantly, multi-antenna IoT devices allow for development of sophisticated signal processing solutions in A-IoT systems, such as the Space-Time Code (STC). For example, orthogonal Space-Time Block Code (STBC) can provide full spatial diversity with simple design and linear decoding complexity. It can be easily integrated into an A-IoT system with multiple antennas at both the BS and device. Depending on whether CSI is available, a coherent or non-coherent receiver with different levels of complexity and performance can be

applied [12]. However, using multiple antennas in A-IoT systems may increase power consumption and implementation complexity. This poses significant challenges, particularly for the devices, which must meet stringent energy and cost requirements.

B. MULTI-NODE JOINT TRANSMISSION

For the topology discussed in Section II, there may be scenarios where multiple nodes simultaneously communicate with multiple devices. In this case, it is necessary to study how to design new transmission strategies to improve the effective distance of data or energy transfer. In general, the joint transmission of multiple BSs has been widely used to enhance the performance of mobile cellular systems. Naturally, the similar principle can also be applied to A-IoT systems, especially for indoor scenarios. The distance between the BSs is usually only tens of meters, and the transmission power of each individual BS is also limited. Therefore, multiple BSs around a given A-IoT device can simultaneously send energy signals. This helps increase the communication distance between the device and the BS, thereby increasing the coverage range of the downlink. However, due to the complexity and energy constraints of A-IoT devices, coherent detection becomes very challenging on the downlink. On the other hand, the uplink signal from the same A-IoT device can be received simultaneously at the neighboring multiple BSs, where spatial channel diversity benefits can be exploited with the joint detection technique to improve the performance. Therefore, developing a multi-node joint transmission strategy for A-IoT systems is important but extremely challenging.

C. COEXISTENCE WITH EXISTING NETWORKS

When RF signals are used to transmit energy, such as using dedicated continuous waves for energy transfer, inter-band/cell interference generally occurs. In such case, if the sole approach to improve the quality of the backscattering signal is to increase the transmission power, eliminating out-of-band emissions that interfere with signals from other systems will become extremely difficult. Thus, mobile network providers are unable to rapidly deploy A-IoT networks in existing BSs with merely software updates. Instead, it is necessary to update or replace existing hardware in BSs. Furthermore, new signal algorithms are required to efficiently handle interference at various layers. The functionality of A-IoT devices, on the other hand, which is typically limited by size and cost, is another key factor impacting the quick deployment within existing networks. As a result, achieving effective coexistence between A-IoT systems and other existing wireless networks is a significant challenge from the perspective of network deployment.

D. DEVICE LOCALIZATION

Given the unique characteristics of backscatter technology and the limitations of device capabilities, determining

the relative or absolute location of A-IoT devices poses significant challenges. Consequently, it is more feasible to prioritize less complex localization approaches from a point of view of practicality. As we know, backscatter channel has significantly different fading characteristics compared with conventional wireless links. Specifically, the emitter-to-tag channel is the product of the forward and backscatter channels, also known as a dyadic channel. Toward this end, it makes channel estimation and acquisition much more complicated resulting in increased burdens to obtain accurate localization. Furthermore, the A-IoT device may necessitate the accumulation of sufficient energy to facilitate responses to activation signals sent by the BS. However, since the BS is unaware of the current energy status of device, it cannot estimate the time required for the devices to energise before responding. As a result, determining the individual contributions of propagation delay and charge delay to total delay is difficult for the network, making accurate positioning estimation based on propagation delay problematic.

E. SPECTRUM ALLOCATION

A-IoT systems may operate in either TDD or FDD modes, each with distinct radio spectrum requirements. Therefore, the selection of frequency bands should remain flexible, encompassing both unlicensed and licensed options. When operating in unlicensed frequency bands, A-IoT systems must adhere to specific requirements. For instance, devices operating in the unlicensed bands must implement channel access schemes like Listen-Before-Talk (LBT), and are subject to maximum transmission power limits based on frequency regulations. For licensed frequency bands, considering the characteristics of backscatter communication and the goal of achieving long communication distances, A-IoT systems are more likely to be deployed in sub-6 GHz bands. However, finding suitable bands within the already crowded sub-6 GHz spectrum will present a significant challenge for the industry.

F. RADIO RESOURCE MANAGEMENT

Effectively utilizing limited radio resources to serve multiple IoT devices is another significant challenge for A-IoT systems. Given the massive number of A-IoT devices with small data load, traditional radio resource management that prioritize maximizing system throughput will no longer be suitable. Instead, resource allocation schemes that balance performance and fairness will become a key focus of future research. Moreover, since a plenty of A-IoT applications are delay-sensitive, resource management must be optimized using metrics that account for delay, such as Age-of-Information (AoI). In practical communication scenarios, the limitations of A-IoT devices in terms of complexity and power consumption make it difficult for these devices to transmit pilot signals or provide feedback on channel state information (CSI). As a result, inaccurate CSI estimation is inevitable in A-IoT systems, which will place higher demands on the design of resource management schemes.

G. SECURITY

Two of the most important security concerns for A-IoT systems are data confidentiality and authentication. It is crucial to ensure that devices respond only to legitimate requests made under authorized access and to prevent eavesdropping or interference with data. However, the limited processing capabilities and energy resources of A-IoT devices make it difficult in achieving these goals. Therefore, it is advisable to offload security processing tasks with high complexity to the base station, which typically has larger computational capabilities. This approach aims to strike a balance between security and resource constraints. Moreover, it is essential to investigate lightweight cryptographic algorithms designed specifically for resource-constrained devices, e.g., the use of lightweight block ciphers or hash functions optimized for efficiency. Additionally, one possible solution for Device A may involve the use of pre-shared keys between A-IoT device and BS, providing a certain level of security. As we know, security is usually about finding the balance between the required level of protection and the limitations of the system. The selection of specific security measures should align with use cases and perceived risks. With technological advancements, there may be ongoing developments in A-IoT security tailored for resource-constrained systems.

VI. CONCLUSION

A-IoT systems heavily rely on backscatter communication technology. More importantly, systematic study and development are required for A-IoT to mature into a viable future system. First of all, this article has thoroughly examined and presented the ongoing A-IoT standardization initiatives in 3GPP. Following that, the essential A-IoT technologies associated to the 3GPP technical framework have been addressed. Furthermore, we have further implemented a comprehensive A-IoT demonstration system. The feasibility of A-IoT has been well demonstrated by field experimental results achieved by this system, e.g., it can support a communication rate of 1 Mbps at a distance of more than 10 meter. Numerous challenges persist regarding the network operation and performance improvement of A-IoT systems. This paper has undertaken a preliminary study on these challenges, recognizing that there are more issues yet to be addressed.

REFERENCES

- [1] Ericsson. (Jun. 2023). *Ericsson Mobility Report*. [Online]. Available: <https://www.ericsson.com/en/reports-and-papers/mobility-report/reports/june-2023>
- [2] K. Zheng, F. Hu, W. Wang, W. Xiang, and M. Dohler, "Radio resource allocation in LTE-advanced cellular networks with M2M communications," *IEEE Commun. Mag.*, vol. 50, no. 7, pp. 184–192, Jul. 2012.
- [3] X. Xiong, K. Zheng, L. Lei, and L. Hou, "Resource allocation based on deep reinforcement learning in IoT edge computing," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 6, pp. 1133–1146, Jun. 2020.
- [4] G. A. Medina-Acosta, L. Zhang, J. Chen, K. Uesaka, Y. Wang, O. Lundqvist, and J. Bergman, "3GPP release-17 physical layer enhancements for LTE-M and NB-IoT," *IEEE Commun. Standards Mag.*, vol. 6, no. 4, pp. 80–86, Dec. 2022.

- [5] L. Mohjazi, B. Selim, M. Tatipamula, and M. A. Imran, "The journey toward 6G: A digital and societal revolution in the making," *IEEE Internet Things Mag.*, vol. 7, no. 2, pp. 119–128, Mar. 2024.
- [6] H. Tataria, M. Shafi, A. F. Molisch, M. Dohler, H. Sjöland, and F. Tufvesson, "6G wireless systems: Vision, requirements, challenges, insights, and opportunities," *Proc. IEEE*, vol. 109, no. 7, pp. 1166–1199, Jul. 2021.
- [7] M. Na, J. Lee, G. Choi, T. Yu, J. Choi, J. Lee, and S. Bahk, "Operator's perspective on 6G: 6G services, vision, and spectrum," *IEEE Commun. Mag.*, vol. 62, no. 8, pp. 178–184, Aug. 2024.
- [8] D. C. Nguyen, M. Ding, P. N. Pathirana, A. Seneviratne, J. Li, D. Niyato, O. Dobre, and H. V. Poor, "6G Internet of Things: A comprehensive survey," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 359–383, Jan. 2022.
- [9] V. Liu, A. Parks, V. Talla, S. Gollakota, D. Wetherall, and J. R. Smith, "Ambient backscatter: Wireless communication out of thin air," *ACM SIGCOMM*, vol. 43, no. 4, pp. 39–50, Aug. 2013.
- [10] G. Yang, Q. Zhang, and Y.-C. Liang, "Cooperative ambient backscatter communications for green Internet-of-Things," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 1116–1130, Apr. 2018.
- [11] G. Vougioukas and A. Bletsas, "Switching frequency techniques for universal ambient backscatter networking," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 2, pp. 464–477, Feb. 2019.
- [12] W. Liu, S. Shen, D. H. K. Tsang, and R. Murch, "Enhancing ambient backscatter communication utilizing coherent and non-coherent space-time codes," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6884–6897, Oct. 2021.
- [13] N. Van Huynh, D. T. Hoang, X. Lu, D. Niyato, P. Wang, and D. I. Kim, "Ambient backscatter communications: A contemporary survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2889–2922, 4th Quart., 2018.
- [14] *Study on Ambient IoT (Internet of Things) in RAN (Release 18)*, document 3GPP TR 38.848 V18.0.0, Jun. 2023.
- [15] M. M. Butt, N. R. Mangalvedhe, N. K. Pratas, J. Harrebek, J. Kimionis, M. Tayyab, O.-E. Barbu, R. Ratasuk, and B. Vejlgaard, "Ambient IoT: A missing link in 3GPP IoT devices landscape," *IEEE Internet Things Mag.*, vol. 7, no. 2, pp. 85–92, Mar. 2024.
- [16] *Study on Solutions for Ambient IoT (Internet of Things) RAN (Release 19)*, document 3GPP TR 38.769 V0.0.1, Feb. 2024.
- [17] F. Rezaei, D. Galappaththige, C. Tellambura, and S. Herath, "Coding techniques for backscatter communications—A contemporary survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 1020–1058, 2nd Quart., 2023.
- [18] Y. Tao, B. Li, C. Zhao, and Y.-C. Liang, "Hardware-efficient signal detection for ambient backscattering communications," *IEEE Commun. Lett.*, vol. 23, no. 12, pp. 2196–2199, Dec. 2019.
- [19] Y. Chen and W. Feng, "Novel signal detectors for ambient backscatter communications in Internet of Things applications," *IEEE Internet Things J.*, vol. 11, no. 3, pp. 5388–5400, Feb. 2024.
- [20] N. Mi, X. Zhang, X. He, J. Xiong, M. Xiao, X.-Y. Li, and P. Yang, "CBMA: Coded-backscatter multiple access," in *Proc. IEEE 39th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Jul. 2019, pp. 799–809.
- [21] J. Guo, X. Zhou, S. Durrani, and H. Yanikomeroglu, "Design of non-orthogonal multiple access enhanced backscatter communication," *IEEE Trans. Wireless Commun.*, vol. 17, no. 10, pp. 6837–6852, Oct. 2018.



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